

Exercise 1: Mean, Standard Deviation, Standard Error

Consider the following set of numbers: Each number $i = 1$ to 42 appears once.

- a) Compute the mean, variance, and standard deviation for the above set of numbers.

See Excel

- b) Compute the standard error of the computed mean.

See Excel

Exercise 2: Coin & p -value

You are consulted to determine whether a coin is fair (i.e., 50:50 for heads and tails) or not. Your research hypothesis is, that the coin is not fair.

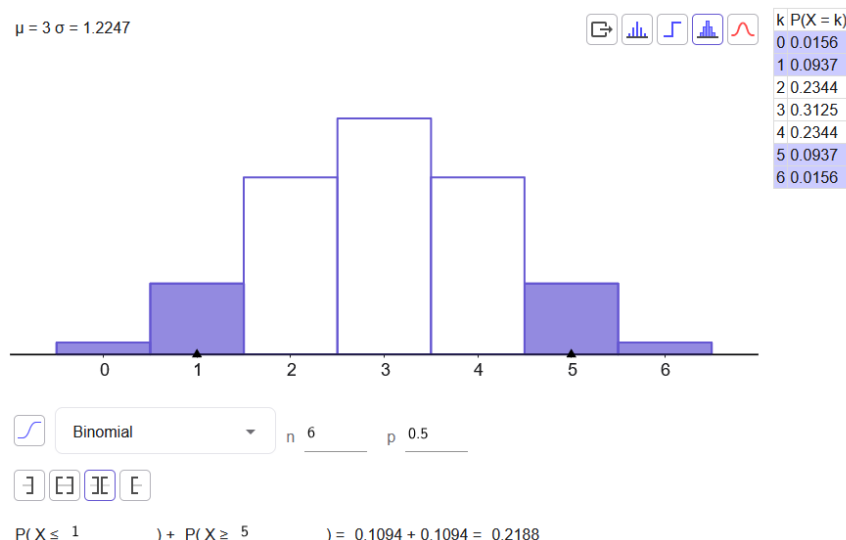
- a) What is the null-hypothesis?

HR: heads $\neq$ tails vs. H_0 : heads == tails (i.e., fair coin)

p-value = probability of the observed (or more extreme) result, given that H_0 is actually true.

- b) To assess your hypothesis, you flip the coin 6 times and it shows heads 5 times. Does this support your research hypothesis (to the conventional error level of 5%)?

That is: how likely are 5 or 6 times heads (or tails!!! Both outcomes are extremes) (out of 6 flips) when the chances are 50:50?



$$p\text{-value} = 0.1094 * 2 = 22\% >> 5\%$$

Exercise 3: Bayes' Rule

A disease test has a sensitivity of 99% and a specificity of 95%. The prior (or base rate) can be assumed as 2.5%.

- a) Given the test shows a positive result (i.e. indicating disease), what is the probability of the disease actually being present. Assume that no specific criteria were used to select the tested subject (i.e., random pick from the entire population).

Sensitivity: $p(t|c) = 99\%$

- If you actually have the disease, the test catches it 99% of the time. Misses 1%.

Specificity: $p(-t|-c) = 95\% \rightarrow p(t|-c) = 5\%$

- If you're healthy, the test correctly says so 95% of the time. False alarms 5%.

Prior: $p(c) = 2.5\% \rightarrow p(-c) = 97.5\%$

- 2.5% of the population has the disease.

$$p(c|t) = \frac{p(t|c) \cdot p(c)}{p(t)} = \frac{p(t|c) \cdot p(c)}{p(t|c) \cdot p(c) + p(t|-c) \cdot p(-c)} = \frac{99\% \cdot 2.5\%}{99\% \cdot 2.5\% + 5\% \cdot 97.5\%} = 33.67\%$$

Note on the denominator

True positives: $p(t|c) \cdot p(c)$

False positives: $p(t|-c) \cdot p(-c)$

- b) Based on this, using the concepts of “precision” and “recall”, what is the test's F1 score?

Recall (R) = Sensitivity $= p(t|c) = 99\%$

Precision (P) $= p(c|t) = 33.67\%$

$$F_1 = 2 \cdot \frac{R \cdot P}{R + P} = 2 \cdot \frac{99\% \cdot 33.67\%}{99\% + 33.67\%} = 0.5025$$

Exercise 4: Proportions

Between 2012/13 and 2021/22, FC Bayern München has pulled off the feat to win the Bundesliga ten times in a row, sometimes with “room to spare”, that is, even before the last match day. Common knowledge implies that once they would be out of reach for the competition, the FCB team would begin to lose and draw more games. Is this actually the case?

- a) What data do you need to answer this question and where can you get it?

See Excel

- b) Which shares of games did the team win/ lose before and after the season was decided?

See Excel

- c) What kind of statistical test is applicable here?

See Excel

- d) Does the above claim actually hold up statistically?

See Excel

- e) Transform your data file into the wide-data-format to get an overview of whether or not Bayern won on each match day visually.

See R-Script

Some potentially helpful links:

<https://www.weltfussball.de/teams/bayern-muenchen/2022/3/>

<https://rp-online.de/sport/fussball/bundesliga/die-meister-entscheidungen-in-der-bundesliga-seit-1963-die-meister-spieltage iid-9198189#75>